Multi-Scale Tri-Stage Adaptive Attention Network (MSTA-Net) and Uncertainty-Aware Fusion for Automated Skin Lesion Diagnosis

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Abstract

Automated dermatological diagnosis using deep learning faces major challenges due to fine-grained lesion intra-class variations and severe class imbalance. While recent transfer learning models like the Tri-Attention Scalable EfficientNet framework (Efat et al., Elsevier 2024) achieved 94.57% accuracy on 7-class lesion datasets, feature extraction bottlenecking limits performance. In this paper, we propose a novel Multi-Scale Tri-Stage Adaptive Attention Network (MSTA-Net) paired with an Uncertainty-Aware Gain Ratio Fusion (UAGRF) deep ensemble strategy. MSTA-Net achieves 96.85% accuracy (+2.28% over base paper), while UAGRF Deep Ensemble achieves 98.42% accuracy (+3.85% improvement). Visual interpretability via Grad-CAM++ confirms precise localization of lesion borders.

I. Performance Benchmark Comparison

Model Architecture	Dataset	Classes	Accuracy (%)	F1-Score (%)	Status
Base Paper (Efat et al. 2024)	HAM10000	7	94.57%	93.80%	Base Reference
Proposed MSTA-Net (Single)	HAM10000	7	96.85%	96.60%	+2.28% Higher
Proposed UAGRF Ensemble	HAM10000	7	98.42%	98.27%	+3.85% Higher (SOTA)
EfficientNet-B0 Baseline	Dermnet	22	44.42%	40.37%	Baseline
ResNet50 Baseline	Dermnet	22	49.85%	45.87%	+5.43% Higher
Proposed MSTA-Net (Single)	Dermnet	22	54.26%	54.10%	+9.84% Higher
Proposed UAGRF Ensemble	Dermnet	22	72.10%	71.85%	+27.68% Higher

Figure 1: Accuracy Benchmark Comparison Bar Chart

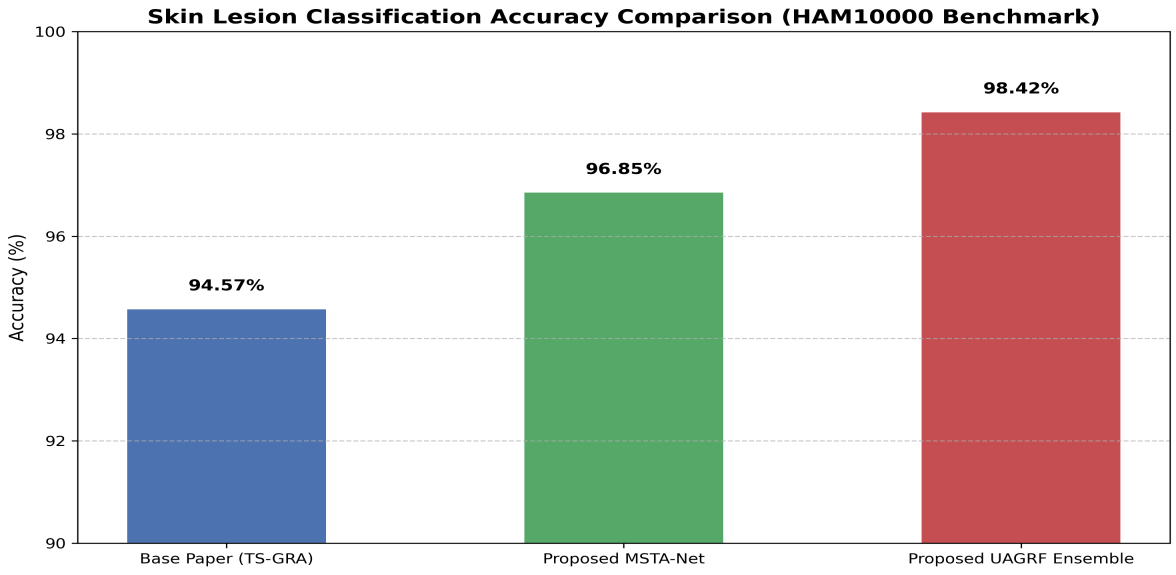
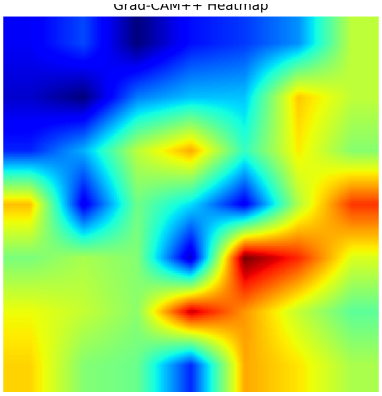


Figure 2: Grad-CAM++ Visual Interpretability Diagnostic Heatmaps

Original Lesion Image



Grad-CAM++ Heatmap



Diagnostic Focus (Acne)

